```
      "stats": {                          <-- The stats query returns all
        "field": "deaths"                     five core metrics at once.
      }
    }
  }
}
```

This `stats` query returns `count`, `avg`, `max`, `min`, and `sum`, all in one shot. Here's the response from our initial `stats` query:

```
"aggregations" : {
  "all_stats" : {
    "count" : 10,
    "min" : 30772.0,
    "max" : 561142.0,
    "avg" : 163689.1,
    "sum" : 1636891.0
  }
}
```

> **NOTE** If you are curious, swap the `stats` query with an `extended_stats` query and check the result. You'll see a lot more stats, including variance, standard deviation, and others. I'll leave it to you to experiment (the code is available with the book's files).

Now that we know what metric aggregations are all about, let's briefly look at another type of aggregations: bucket aggregations.

2.6.2 Bucket aggregations

A bucket aggregation (or simply *bucketing*) segregates data into various groups or *buckets*. For example, we can add groups like these to our buckets: a survey of adult groups by age bracket (20–30, 31–40, 41–50), movies by review ratings, or number of new houses constructed per month.

Elasticsearch provides at least two dozen aggregations out of the box, each having its own bucketing strategy. What's more, we can nest aggregations under the main buckets. Let's look at a couple of bucketing aggregations in action.

HISTOGRAM BUCKETS

The `histogram` bucketing aggregation creates a list of buckets representing a numerical value by going over all the documents. For example, if we want to categorize countries by the number of critical patients in buckets of 2,500, we can write the following query.

Listing 2.29 Countries by critical patients in buckets of 2,500

```
GET covid/_search
{
  "size": 0,
  "aggs": {
    "critical_patients_as_histogram": {     <-- User-defined name
                                                of the report
```